

RUCHI SINHA

Full Stack Engineer · AI/GenAI Integration Specialist

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with **5+ years** of experience building scalable web applications and **2+ years** of hands-on AI/GenAI integration experience. Specialised in designing and delivering **RAG pipelines**, **LLM-powered assistants**, **voice AI agents**, and **multi-tenant SaaS platforms** using React.js, Next.js, Node.js, and PostgreSQL/MongoDB. Proven expertise in semantic search, prompt engineering, embedding generation, and conversational AI — combined with a strong backend foundation in RESTful APIs, real-time communication (Socket.IO), RBAC, and cloud services (AWS S3).

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend: React.js, Next.js, Redux, Redux Toolkit, Redux-Saga, Tailwind CSS

Backend: Node.js, Express.js, RESTful API Design, Event-Driven Architecture

Databases: PostgreSQL, MongoDB, Mongoose ODM, Prisma ORM, pgvector

AI / GenAI: RAG Pipelines, OpenAI API, LLM Prompt Engineering, Semantic Search, Text Embeddings, Vector Databases, AI Guardrails, VAPI (Voice AI), Conversational AI

System Design: Multi-Tenant Architecture, RBAC, Microservices, Distributed Systems, API Design

Real-Time: Socket.IO, WebSockets, Event-Driven Architecture

Cloud & DevOps: AWS S3, Git, GitHub, Postman

Performance: Memoization, Lazy Loading, Code Splitting, Async Processing, Query Optimization

WORK EXPERIENCE

Software Engineer

Oct 2025 – May 2026

Hestabit Technologies

Noida, India

- Architected an **AI Voice Agent & Campaign Platform** (SaaS) designed to deploy automated conversational AI agents across **3+ business verticals** (healthcare, retail, feedback collection) at scale.
- Designed a **campaign orchestration engine** architected to support bulk outbound calls to **500+ contacts per campaign**, with retry logic, cron-based scheduling, and real-time call tracking.
- Engineered a **batch processing pipeline** with controlled concurrency management for sequential VAPI dispatch — preventing API overload and ensuring campaign reliability at scale.
- Implemented **dynamic prompt engineering and context injection** — configuring each AI assistant with company-specific context, campaign objectives, and conversation flows tailored per business vertical.
- Integrated **VAPI** for real-time conversational voice AI and **Twilio** for telephony, call routing, and result processing across multi-tenant campaigns.
- Built scalable Node.js/MongoDB backend handling agent workflows, session lifecycle management, and tenant-isolated data storage with RBAC enforcement.
- Implemented **Socket.IO event-driven architecture** for real-time monitoring of active calls and system state transitions.
- Reduced API latency by **25%** through optimised asynchronous processing and elimination of redundant database queries.

Software Engineer

Jul 2023 – Oct 2025

Chetu India Pvt. Ltd

Noida, India

- Designed and delivered an **AI-Powered Insurance Policy Assistant** using RAG — processing **20+ policy documents** across **5+ enterprise tenants**, enabling **200+ employee queries** via a self-service AI chat interface.
- Built the **document ingestion pipeline**: extracted text from PDF policy documents, applied chunking strategies, generated vector embeddings via OpenAI API, and stored chunks with metadata in PostgreSQL using pgvector.
- Implemented **semantic similarity search** — converting user queries into embeddings and retrieving contextually relevant policy chunks, resolving sequential embedding generation bottleneck via async background processing.
- Engineered **RAG-based prompt construction** injecting only retrieved chunks into LLM context — reducing token usage, improving answer accuracy, and minimising hallucinations.
- Applied **AI guardrails via prompt engineering** to restrict responses strictly to policy content and reject out-of-scope queries; delivered **source citations** in every AI response for trust and explainability.

- Developed role-specific dashboards for a **Supply Chain Management System** (Next.js) and a **real-time multiplayer gaming platform (Mega2)** (React.js) — both with granular RBAC, Socket.IO real-time features, and Redux state management.
- Optimised performance across both platforms using memoization, lazy loading, and code splitting — reducing page load times on inventory dashboards and live game sessions.

Software Developer

May 2021 – Jul 2023

Integrated Personnel Connecting Tech India

- Developed and maintained RESTful APIs using Node.js and MongoDB for an **HRMS platform**, covering authentication, authorisation, and employee lifecycle workflows.
- Built a Next.js marketing website using static site generation (SSG) for fast load times and improved search engine visibility.
- Integrated **AWS S3** for secure, scalable document storage; developed React.js frontend modules with Redux-Saga for real-time data updates.
- Designed and optimised MongoDB schemas for performance and data integrity across core HR modules.

EDUCATION

Master of Computer Applications (MCA)

2018 – 2021

Patliputra University

Bihar, India

- CGPA: 8.86 / 10

Bachelor of Computer Applications (BCA)

2015 – 2018

Aryabhatta Knowledge University

Bihar, India

- CGPA: 9.14 / 10

KEY ACHIEVEMENTS

25% reduction in API latency at Hestabit via async processing and query optimisation.

Architected a **production RAG pipeline** end-to-end — PDF ingestion, chunking, OpenAI embeddings, pgvector semantic search, and LLM answer generation with citations — serving 5+ enterprise tenants.

Resolved **sequential embedding bottleneck** in RAG system by implementing async background processing, unblocking document ingestion at scale.

Delivered **AI Voice Agent SaaS platform** architected for 500+ contacts per campaign with dynamic per-tenant prompt engineering across 3+ business verticals.

Consistently high academic performance — **8.86 GPA (MCA)** and **9.14 GPA (BCA)**.